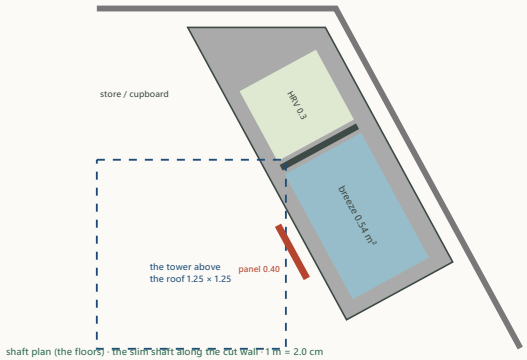
Air system study v3 · one tower, two separate air paths · Taif / Palm House V7.1

Owner goals 1 Nov 2026 (design review E-01) and the owner's decisions on the open items (v3.1: assist fan, bird mesh, the HRV base under the tower, the condensers south, the make-up inlet) · a gentle natural breeze for the courtyard seating in pleasant weather, and the fresh-air intake of the house's ventilation unit all year · no water in the air system, no pads, no earth pipes, no human entry · **every size and performance figure is ASSUMED, to be designed by the MEP engineer** compiling this study is not engineering approval

1 · The concept

- **One tower** (V9.3, owner): **a slim shaft hugging the cut wall** (1.58 m² on walls, **0.9 m² clear**, 0.575 × 1.565 m, assumed) rising to **the malqaf tower** set back on the attic's corner, the symbol of the house: a 1.25 m square in honey stone, a plain shaft, a cornice band, a head of 4 slender openings per face each with a small semicircular top, a flat Hijazi crowning cornice, 4.88 m above the roof finish (crown +12.18, 3.5 m above the attic coping); the one element above the attic (`studies/roof.typ`, `studies/malqaf_tower.typ`).
- **Path (a), the courtyard breeze:** a four-sided head → a breeze shaft → a damper → a shallow channel in the ground slab → the grille under the outdoor TV: a gentle natural breeze for the seating in pleasant weather; not cooling (if it is too hot the family does not sit there); no water, no pads, no earth pipe; a quiet **assist fan** in the plenum behind the grille, on its own switch (owner, 1 Nov 2026; § 4).
- **Path (b), the house's fresh air:** an airtight intake compartment in the tower with its own louvre (the west face, just below the band) and an accessible pre-filter, feeding the ventilation unit in its **plant room behind the attic** (§ 2, never seen); the supply and extract risers to the ground floor run down the slim shaft's HRV compartment, the first floor is fed straight through the roof slab. The two paths are airtight from each other.
- **Low maintenance, no entry:** every routine task is done from the roof (the bird-mesh frames lowered on cords to the **roof hatch** 0.50 m in the tower's south face behind the attic, the flaps, the top damper, the pre-filter), at the plant room's doors (the unit, the filters, the core), from the **access panel 0.40 × 0.40 at the shaft's foot in the maid bath** (the sump), from the laundry and kitchen floors (the channel's covers) or from the courtyard (the grille and the fan box). Nothing opens into bedroom 1; no shaft, pit or buried space is entered; the shaft is smooth-rendered inside (§ 7).

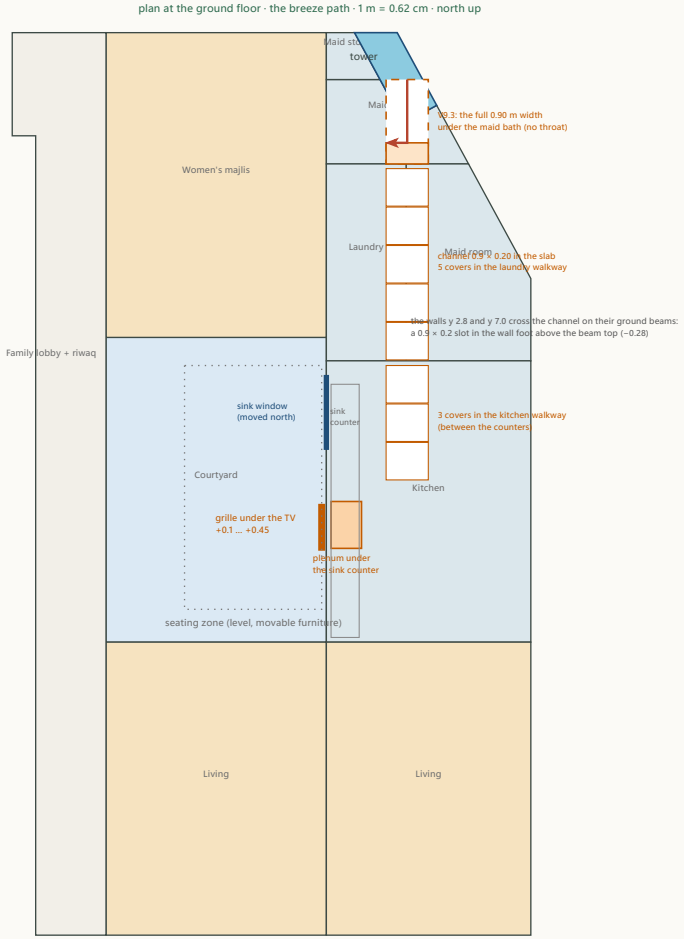


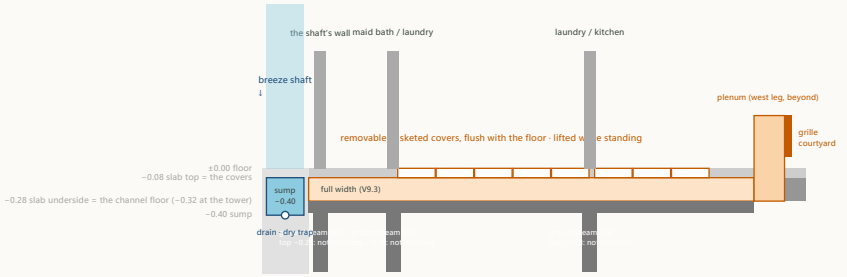
Air system study v3 · checks passed (9)

2 · The tower

- **The slim shaft, 0.9 m² clear** (assumed, inside the walls): 0.575 m deep between the cut wall and one new 200 mm wall parallel to it, 1.565 m long, a square end wall, the tip a solid corner pier; split by an airtight partition (0.1 m, square to the cut) into the **breeze shaft 0.54 m²** (south part) and the **HRV compartment 0.3 m²** (north part: the two Ø250 risers); the maid bath and bedroom 1 each get **one straight wall parallel to the cut** and a square return, the store and the cupboard north of the y 1.0 partition.
- **Head (V9.3, head (c)), four faces, in the house's style:** 4 slender openings per face (+10.35 … +11.73, each 0.14 m wide with a small semicircular top, honey-stone ribs; 3.06 m² free in all). **The problem checked:** an undivided head open on four sides short-circuits (the wind enters windward and leaves leeward without reaching the courtyard). **The simplest arrangement that works:** the head only is divided into four quarters by a thin + divider; at the foot of each quarter a **lightweight back-draught flap** closes under the leeward suction and opens under the windward pressure, so only the windward quarter(s) feed the single shaft below. **Its sizing and the damping are the MEP engineer's.**
- **No louvres; a bird mesh, cleanable from the roof:** the slender slots are the openings (rain falls to the shaft's sump); behind them a **wide 20 mm bird mesh only** (≈ 90 % free; no insect mesh on the breeze path: owner, 1 Nov 2026) in **frames lowered on cords** inside each quarter to the roof hatch and brushed / vacuumed there; no ladder on the tower, no entry.
- **Damper:** a motorised blade damper across the breeze path at the tower's foot (just above the plenum, ≈ +7.9), a standard **spring-return HVAC actuator** (closes on power loss; ≈ 5 W holding, 24 V), on a wall switch beside the seating (by the TV) with an "open" light; reached from the **roof hatch 0.50 m in the tower's south face** (behind the attic, +7.6 … +8.1): no entry.
- **ERV intake compartment:** a strip of the tower's section (0.27 m², the west side) from the head down to the roof, with its own louvre on the tower's west face just below the band (+9.63 … +10.03, a bird mesh) and a slide-in **pre-filter** (ISO Coarse 60 %) at its foot, pulled out through the roof hatch; the Ø315 intake duct leaves the tower's foot to the plant room (0.6 m, a low duct box on the roof behind the attic).
- **The plant room (V9.3; the wide base under the tower removed):** behind the attic, x 8.7–10.3, y 2.9–4.3 (1.6 × 1.4 m outside), its top +8.6 below the attic coping (+8.68): **never seen**; inside ≈ 1.68 m² and 1.18 m clear: a **low service compartment**, not a room (the unit on a 0.10 plinth, its front at a pair of doors on the south face, filters and the core pulled out on the roof walkway); the exhaust led along the attic to a low grille in its cut wall (+7.7), 4.7 m from the intake.

3 · The courtyard breeze path: the channel in the slab, its covers, the plenum and the grille · plan and long section





long section along the channel (x 10.2), looking east, heights × 2.5 · 1 m = 0.62 cm · the channel never leaves the slab depth

- **Route:** from the breeze shaft's foot (the tower's south wall, y 1.0) **straight south on x 10.67** under the maid bath (y 1.0–2.8), the laundry walkway (2.8–7.0) and the kitchen walkway between the counters (7.0–10.5), then **one 90° bend west** under the sink counter (y 10.0–11.0) into the plenum in the wall base and the grille: axis 10.35 m, one bend. The slot through each of the three walls (y 1.0, 2.8, 7.0) is 0.9 × 0.20 in the wall foot **above the ground beam's top (−0.28)**: no beam, footing or floor is cut or lowered.
- **Size and levels:** clear 0.9 × 0.2 m (0.18 m²) from the slab top (−0.08, the covers flush with the finished floor) to the slab underside (−0.28), its own base slab 0.10 below; V9.3: **the full width under the maid bath too** (no throat; the bath's WC and shower drains stay off the line: the bathrooms interior study): 0.18 m², the controlling section. The floor falls 0.5 % back to the tower's **sump** (−0.40, a Ø75 drain with a waterless dry trap against odours, to the east yard): rain that passes the louvres and any condensation leave at the tower, never towards the courtyard.
- **Access without entry:** 8 removable sealed covers (0.9 × 0.8, stainless trays stone-faced like the floor, gasketed against wash-water and insects, lifted with two suction cups) in the laundry and kitchen walkways; the section under the maid bath and the sump are reached with a flat brush and a vacuum rod from the first cover and from the shaft's foot; the plenum and the west leg (0.65 m) from the courtyard with the grille removed. The channel's surfaces smooth (a sealed epoxy render), the corners coved.
- **Outlet:** the decorative grille under the TV, 1 × 0.35 m at +0.1 … +0.45 (0.25 above the seating floor), with a **manually adjustable louvre** that directs the air along the floor to the benches, clear of the cushions (no bench on the east side) and 1.65 m from the fountain enclosure's splash; the grille lifts off for cleaning.

4 · The breeze, estimated honestly · 5 · The house ventilation (the sealed house)

4 · What the courtyard will feel

The path's resistance, referred to a 0.10 m² section (the V9.2 throat, so the figures compare; V9.3 the courtyard flow ≈ 1.37 × V9.2's) (**assumed loss coefficients, ASHRAE Fundamentals ch. 21 orders of magnitude**):

Element	ζ (0.10 m²)
the head: 4 slots per face, bird mesh (4 faces, 3.06 m² free)	0
back-draught flaps and the shaft (0.54 m²)	0.03
damper, open (in the tower's foot)	0.02
the plenum's two bends at the roof (0.60 m²)	0.03
the turn from the shaft's foot into the channel	0.15
channel friction, 10.3 m, 0.18 m² (full width, no throat)	0.19
the bend and the plenum	0.25
the grille (0.35 m², 50 % free)	0.82
Total	**1.5**

Driving pressure: wind at the head ½ρv²·ΔCp with ρ 1.15 kg/m³ (Taif, 1 700 m) and ΔCp 0.8 (windward face against the courtyard well, assumed); in still air the stack of the 12 m path with 3 K difference: 1.35 Pa.

Wind at the head	Δp	flow	grille	felt at the benches	What it is
weak, 3 m/s	4.1 Pa	0.22 m³/s (789 m³/h)	1.25 m/s	≈ 0.35 m/s	barely perceptible: a faint stirring at the floor
moderate, 6 m/s	16.6 Pa	0.44 m³/s (1578 m³/h)	2.5 m/s	≈ 0.7 m/s	a gentle, pleasant breeze on the legs and hands
still (stack only)	1.35 Pa	0.13 m³/s	0.72 m/s	≈ 0.2 m/s	nothing felt

- **Honest conclusion:** the breeze is perceptible only with a **moderate wind at the head** (≈ 5 m/s and more at 9 m above the ground); Taif's spring and summer afternoons often have it, calm evenings do not. In still weather the tower gives nothing the family will notice. The path is as short and open as the house allows (V9.3: the throat under the maid bath removed, the channel at its full width: ≈ 1.37 × the V9.2 flow); the head's screens are not the limit.
- **Assist fan (owner decision, 1 Nov 2026):** a quiet **EC extract-type fan** in the breeze path, pulling the air down from the tower to the grille: a flat fan box (two Ø250 EC axial fans at low speed, ≈ 0.9 × 0.3 × 0.3) in the **plenum directly behind the grille**, 0.15–0.25 m³/s, ≤ 30 dB(A) at the benches, 40–60 W, on its **own switch** beside the seating (separate from the damper switch; the fan runs only with the damper open: an interlock): it gives the moderate-wind breeze on any evening. **Service without any confined space:** the grille lifts off and the box slides out through the grille opening (1.0 × 0.35) into the courtyard. **The MEP engineer sizes it and its sound.** Wind data for Taif (the airport station) to be checked; the figures above are a concept estimate, not a design.

5 · The house's ventilation

- **One unit** for the whole house in the **plant room behind the attic** (§ 2; `studies/roof.typ` Z2a, V9.3): x 8.7–10.3, y 2.9–4.3, ≈ 1.68 m² inside, 1.18 m clear, the unit right behind the south doors with its front at them (filters and the core pulled out on the roof walkway). **Unit (assumed):** a 750 m³/h class counterflow HRV / ERV (sensible core; enthalpy core optional), on a 0.1 plinth in the tower's base, its front at the doors: 750 m³/h class, ≈ 1.2 × 0.7 × 0.7 m, ≈ 110 kg, 90–350 W.
- **ERV or HRV:** Taif is dry and high (summer 30–35 °C by day with low humidity, winter nights near 5 °C). Moisture recovery gives little here: in summer the outdoor air is dry anyway, in winter it only slows the drying of the indoor air. A **sensible-only HRV** (or a unit whose enthalpy core can be swapped for a sensible one) is the better fit: simpler core, no condensate in summer, the winter heat recovery is what matters. **The MEP engineer decides with the unit's data.**
- **Preliminary airflow** (ASHRAE 62.2-2022 §4.1.1: 0.15 L/s·m² of floor + 3.5 L/s per bedroom + 1): conditioned floor ≈ 462 m² clear (502 m² on axes), 5 bedrooms → **90 L/s = 325 m³/h** continuous. **Guest gatherings:** 24 men in the majlis (47 m²) at the ASHRAE 62.1 assembly rate (3.8 L/s·person + 0.3 L/s·m²) add 105 L/s → **boost 196 L/s = 704 m³/h**. The unit above covers it ✓. **Rates to be confirmed against SBC 501 / 1101 and the owner's occupancy.**
- **Intake and exhaust:** the intake from the tower's intake compartment (its louvre on the west face below the band, +9.83); the exhaust led from the plant room along the attic to a low grille in the attic's cut wall (+7.7), blowing east over the lane side, hidden behind the coping. **The AC condensers stay SOUTH** (owner; zone Z2b from y 5.6): 1.3 m clear of the plant room and 4.1 m south of the intake in plan, the intake ≈ 2 m higher than their tops.
- **Filters:** a coarse pre-filter (ISO Coarse 60 %) in the tower compartment and **one fine filter** at the unit's supply, **provisionally ISO ePM1 ≥ 50 %** (**to confirm with the equipment**); a differential-pressure switch or the unit's own filter timer as the service indicator, shown at the plant room door and on the panel.
- **Supply and extract:** supply to every bedroom (1–3, the kids' room, the suite), the living, the family living, the office and the men's majlis (with the boost); extract from every bathroom and WC (the family bath, bath 3, the suite bath and WC, the family WC, the guest WC, the maid bath) and the laundry; transfer air from the rooms to the extract points through **door undercuts of 15–20 mm** or transfer grilles over the bathroom doors, so the system works with the bedroom doors closed.
- **The kitchen hood stays separate:** 750 m³/h out through the east wall; **make-up air** through a filtered, motorised inlet (≈ 300 × 300); **proposed position (the owner wants it discreet):** on the kitchen's east wall at y 12.2–12.5, just under the first-floor stone band (+3.0 … +3.3, in the band's shadow line, a louvre painted the wall colour; inside, the fridge and tall storage stand below it), ducted in the ceiling zone to a diffuser 1.5 m from the hood; interlocked with the hood, so the heavy daily cooking never depressurises the sealed house (a 0.9 m³/h gap would otherwise pull dust through every crack). **The MEP engineer sizes it.**

Intake / exhaust separations (from the plan; the basis in the last column)

Pair	distance	allowance	basis
ERV intake ↔ ERV exhaust	4.7 m	3 m	ASHRAE 62.1-2022 Table 5-1: class 2 exhaust (toilets) → intake ≥ 3 m (10 ft); the exhaust blows east, the intake faces north
ERV intake ↔ kitchen hood outlet (east wall)	12.7 m	4.6 m	class 3 (cooking) → ≥ 4.6 m (15 ft); the hood outlet is 6.6 m below the intake
ERV intake ↔ plumbing vent stack	8.3 m	3 m	vents → ≥ 3 m (10 ft), and the intake 0.9 m above the vent top
ERV intake ↔ LPG cabinet (yard)	13 m	4.6 m	a leak source at ground level: class 3 allowance
ERV intake ↔ generator point (yard)	15.4 m	9 m	class 4 (engine exhaust) → ≥ 9 m (30 ft); the generator's exhaust pipe led away from the house
ERV intake ↔ nearest AC condenser (Z2b)	4.6 m	2 m	not an exhaust: 2 m from the hot plume (manufacturer); the intake is 1 m above the units
ERV exhaust ↔ nearest AC condenser	1.1 m	1 m	stale air into a condenser is harmless; keep 1 m for service

ASHRAE 62.1-2022 §5.5.1 and Table 5-1 (the 3 m / 4.6 m / 9 m figures are the IFC-era "starting allowances" the SBC 501 follows, **not verified**); the gas cabinet and the generator point are site-plan proposals and may move; proposal: keep the condenser zone south of y 5.6 so the intake stays upwind of their plumes in the north-west wind.

Ducts and ceiling heights

- **Vertical route (V9.3):** the first floor's supply and extract straight down from the plant room through the roof slab into the ceiling zone (+6.85) of bedroom 1 and the family part and on to the gallery; the ground floor's through the slim shaft's HRV compartment (two Ø250 risers) into the maid bath / laundry ceiling (+3.0 … +3.3) and on to the kitchen, the dining, the riwaq and the two majlises. Flat ducts in the ceiling zones (assumed); fire dampers where they cross the floor (MEP).

Zone	soffit	duct	ceiling
GF rooms under the 0.20 slabs (women's majlis, riwaq, the wash block, the maid wing)	+3.32	0.1	**+3.17**
GF rooms under the 0.30 ribbed bays (kitchen, dining, men's majlis)	+3.22	0.1	**+3.07**
under beam B1 (women's majlis / kids' wall, soffit +3.20): a 0.3 m band	+3.2	—	**+3.05**
under beam B4 (the riwaq's east wall line, soffit +3.10): a band over the riwaq	+3.1	—	**+2.95**
under beam B5 (the living / dining ceiling step, soffit +3.05): a 0.3 m band at the dining's west edge	+3.05	—	**+2.9**
kitchen (the hood duct 0.25 + the extract duct in the same zone)	+3.22	0.25	**+2.92**
FF rooms under the 0.20 roof slab (bedrooms 2 / 3, the kids' room, the family living, the gallery)	+7	0.1	**+6.85**
FF rooms under the 0.30 roof bays (the suite, the office, the gym)	+6.9	0.1	**+6.75**

- **Finished clear heights:** ≥ +3.07 in the ground-floor rooms, +2.90 only in the 0.3 m band under B5 at the dining's west edge (the living has no ceiling); the first floor ≥ +6.75 (3.15 m clear). **No room gets too low**; the lowest band is at the ceiling step the living already has.

6 · Controls · 7 · Maintenance (no routine task enters a shaft, pit or buried space) · 8 · Decisions and engineering checks still open

6 · Controls

- **Continuous quiet running** at the normal rate (325 m^3/h), **boost** (704) from the kitchen and bathroom switches (timed) and a "guests" button in the hospitality kitchen for the majlis evenings; the fan speeds set at commissioning.
- **The courtyard damper is independent:** its own switch by the seating; spring-return, so it **closes on power loss** and when the switch is off; the **assist fan on its own switch** beside it (interlocked: it runs only with the damper open).
- **Dust-event mode** (a switch at the family entrance or a weather-station input): the unit drops to its **minimum balanced** rate (supply = extract) so the house is never depressurised, the kitchen hood's make-up inlet stays interlocked, the courtyard damper closes; the unit is **not** assumed to recirculate: indoor air cleaning during a storm is **separate equipment** (portable HEPA units in the bedrooms and the living, a proposal).
- **Power failure:** the unit stops, the damper springs shut, the sealed house stays sealed; the alarms keep running on their batteries (`studies/fire_escape.typ`); on restoration the unit restarts at its last setting. The unit may sit on the essentials circuit if the electrical designer agrees (≈ 100–350 W).
- **Indicators:** the filter-service light at the plant room door and on the panel; the damper "open" light at the seating switch.

8 · Decisions (owner) and engineering checks still open

- **Owner (decided 1 Nov 2026; V9.3 8 Nov 2026):** the assist fan yes; bird mesh only; the condensers south of the tower; the make-up inlet discreet (position proposed above); **V9.3: the malqaf tower at the north-east corner (head (c), 3.5 m above the attic, the Hijazi cornice), the slim shaft along the cut wall, the HRV in a plant room behind the attic, the channel at its full width. Still open:** the fan's sound in practice (hear a demonstration unit).
- **MEP engineer:** every rate, size and pressure drop above; the flaps and the damper; the HRV / ERV choice and the unit; the duct sizes and the attenuators; the make-up air; the sump drain and its trap; the controls; the Taif wind data for the breeze.
- **Structural engineer:** the channel in the slab (0.9 × 0.2 within the slab depth, its base slab, the three wall slots above the ground beams), the slim shaft's walls and openings, the tower on the roof slab (≈ 71 kN, the wind moment ≈ 21 kN·m, assumed), the plenum box and the plant room's load (`studies/structure.typ`).
- **3D and other files:** `3d/build_3d.py` / `check_3d.py` still build and check the v2 pit, chamber and duct: to be rewritten for v3 at the next 3D build (`docs/PENDING_CHANGES.md`).

7 · Maintenance (initial intervals; adjust to the manufacturer and the dust)

Component	Task	Reached from	Interval	If neglected
The head's bird mesh (4 quarters)	lower the frames on their cords, brush / vacuum, raise them	the roof hatch in the tower's south face (behind the attic)	6 months; after a dust storm (no fine mesh: little to clog)	the breeze drops; birds enter the shaft
Back-draught flaps (4)	check they swing free, clean the seats	the roof hatch in the tower's south face, by hand	12 months	a leeward quarter stays open: the head short-circuits, no breeze
Courtyard damper and its actuator	cycle it, check the spring return and the switch	the roof hatch, a hinged panel	12 months	the damper sticks open (dust and cold air into the courtyard) or closed
ERV pre-filter (the tower's intake compartment)	slide out, replace	the roof hatch, a second panel	3 months	the unit's fine filter clogs early; less fresh air
Unit filters (the plant room)	replace the fine supply filter and the extract filter	the roof walkway: the plant room's south doors, no entry	3–6 months (the indicator)	less air, more fan power, dust bypass
Heat-recovery core	pull, wash, dry	the roof walkway: through the plant room's doors	12 months	lost recovery, odours
Exhaust louvre and the make-up air inlet	clean the mesh; test the motorised inlet with the hood	the attic's cut wall (behind the attic) / the kitchen wall	6 months	the hood depressurises the house; dust in
Channel covers and channel	lift the trays, vacuum the channel, check the gaskets	the laundry and kitchen walkways, standing	6 months; after a storm	dust blown into the courtyard; wash-water into the channel
Throat section and sump	brush with a rod; check the dry trap, pour water once a year	the first laundry cover and the shaft's foot (no entry)	6 months	odours from the drain; a blocked sump
Plenum and grille	lift off the grille, vacuum the plenum, set the louvre	the courtyard	1 month (dust)	dust on the seating; a blocked grille
Assist fan box (behind the grille)	vacuum the blades, check the bearings and the damper interlock	the courtyard: lift the grille, slide the box out	12 months	noise; less breeze
Supply and extract grilles in the rooms	vacuum	the rooms, a step ladder	6 months	noise, less air

Every task is done from the roof terrace, a room's floor or the courtyard. Nothing requires entering the tower, the channel or any pit: the sump is 0.32 m deep and reached by hand and rod.